Def. A matrix $A \in M_{n\times n}$ is <u>orthogonally diagonalizable</u> if

(*) $A = PDP^{-1}$ where

$P \in M_{n\times n}$ is orthogonal and $(P^T = P^{-1})$

$D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$ is diagonal.

Recall that $P = [u_1 \cdots u_n]$ with columns $u_1, \dots, u_n \in \mathbb{R}^n$ is orthogonal $\Leftrightarrow$ $\{u_1, \dots, u_n\}$ is an orthonormal basis.

Recall also that (*) tells us that the columns of P are eigenvectors of A with

$$A u_j = \lambda_j u_j.$$

Putting these two recollections together we have

A orthogonally diagonalizable $\Leftrightarrow$ P has columns which are an orthonormal basis of $\mathbb{R}^n$ of eigenvectors of A

$\Leftrightarrow$ there is an orthonormal basis of $\mathbb{R}^n$ of eigenvectors for A.

Example 2 in textbook

Take $A = \begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix} \in M_{3\times 3}$, a symmetric matrix.

$$\det(A - xI) = \cdots = -(x-8)(x-6)(x-3)$$

$$\det(A - xI) = 0 \iff x = 8, 6, 3$$

3 distinct eigenvalues in $\mathbb{R}^3 \Rightarrow A$ diagonalizable.

$$\mathrm{Nul}(A - 8I) = \cdots = \mathrm{span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$\mathrm{Nul}(A - 6I) = \cdots = \mathrm{span}\left\{ \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \right\}$$

$$\mathrm{Nul}(A - 3I) = \cdots = \mathrm{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Let's check orthogonality of $\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$

$$\left\langle \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \right\rangle = (-1)^2 + (-1) + 0 = 0.$$

$$\left\langle \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle = (-1) + 1 + 0 = 0$$

$$\left\langle \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \right\rangle = -1 + (-1) + 2 = 0$$

Orthogonal basis !!

We normalize each vector to obtain an orthonormal basis. Set $u_1 = \dfrac{1}{\left\| \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\|} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \dfrac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$.

Similarly $u_2 = \dfrac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$, $u_3 = \dfrac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

$$\Rightarrow \left\{ \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix}, \begin{pmatrix} -1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{pmatrix}, \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix} \right\}$$ is an orthonormal basis of eigenvectors of A.

Let $P = \begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} \in M_{3\times 3}$

Then $A = P \begin{bmatrix} 8 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 3 \end{bmatrix} P^T$ is an orthogonal diagonalization of A.

Theorem 1

Suppose $A \in M_{n\times n}$ is symmetric and $v_1, v_2 \in \mathbb{R}$ are eigenvectors for λ_1 and λ_2 respectively with $\lambda_1 \neq \lambda_2$. Then $\langle v_1, v_2 \rangle = 0$ i.e. $\{v_1, v_2\}$ is orthogonal.

Proof.

$$\begin{aligned}
\lambda_1 \langle v_1, v_2 \rangle = \langle \lambda_1 v_1, v_2 \rangle &= \langle A v_1, v_2 \rangle \\
&= (A v_1)^T v_2 \\
&= v_1^T A^T v_2 \\
&= v_1^T A v_2 \qquad (A^T = A) \\
&= \langle v_1, A v_2 \rangle = \langle v_1, \lambda_2 v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle
\end{aligned}$$

$\Rightarrow \lambda_1 \langle v_1, v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle$

$\Rightarrow (\lambda_1 - \lambda_2) \langle v_1, v_2 \rangle = 0 \qquad \lambda_1 - \lambda_2 \neq 0$

$\Rightarrow \langle v_1, v_2 \rangle = 0$

Theorem 2 [Spectral Theorem Part I]

$A \in M_{n \times n}$ is symmetric $\Leftrightarrow$ A is orthogonally diagonalizable.

Remark. If $A \in M_{n \times n}$ is symmetric with n distinct eigenvalues then Thm 1 applies to the basis of eigenvectors one computes. So the basis of eigenvectors is orthogonal and all you have to do is normalize the basis vectors to get an orthogonal diagonalization.

Example. 3 in textbook.

Take $A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$. A is symmetric.

By Thm 2 we know A is orthogonally diagonalizable.

$\det(A - xI) = \cdots = -(x-7)^2 (x-2)$

Eigenvalues are 7 and 2.

$\mathrm{Nul}(A - 7I) = \cdots = \mathrm{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} \right\}.$

$\left\{ \begin{bmatrix} 1 \\ -2 \\ \end{bmatrix}, \right.$

$$\mathrm{Nul}(A-2I) = \cdots = \mathrm{span}\left\{\begin{bmatrix}-2\\-1\\2\end{bmatrix}\right\}$$

By Thm 1 $\left\langle \begin{bmatrix}-2\\-1\\2\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix}\right\rangle = 0$, $\left\langle \begin{bmatrix}-2\\1\\2\end{bmatrix}, \begin{bmatrix}-1\\2\\0\end{bmatrix}\right\rangle = 0$.

However Thm 1 does <u>not</u> apply to

$$\left\langle \begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}-1\\2\\0\end{bmatrix}\right\rangle = -1+0+0 = -1 \neq 0.$$

Let's use Gram Schmidt to obtain an orthogonal basis for $\mathrm{Nul}(A-7I)$. Set $y_1 = \begin{bmatrix}1\\0\\1\end{bmatrix}$, $y_2 = \begin{bmatrix}-1\\2\\0\end{bmatrix} - \dfrac{\left\langle \begin{bmatrix}-1\\2\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix}\right\rangle}{\left\langle \begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix}\right\rangle}\begin{bmatrix}1\\0\\1\end{bmatrix}$

$$= \begin{bmatrix}-1\\2\\0\end{bmatrix} - \frac{-1}{2}\begin{bmatrix}1\\0\\1\end{bmatrix}$$

$$= \begin{bmatrix}-1\\2\\0\end{bmatrix} + \frac{1}{2}\begin{bmatrix}1\\0\\1\end{bmatrix}$$

$$= \begin{bmatrix}-1/2\\2\\1/2\end{bmatrix}.$$

$\Rightarrow \left\{ \begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}-1/2\\2\\1/2\end{bmatrix}\right\}$ is an orthogonal basis for $\mathrm{Nul}(A-7I)$

$\Rightarrow \left\{ \begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}-1\\4\\1\end{bmatrix}\right\}$ "

Let normalize the vectors in $\left\{ \begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}-1\\4\\1\end{bmatrix}, \begin{bmatrix}-2\\-1\\2\end{bmatrix}\right\}$

the orthogonal basis of eigenvectors.

Let $u_1 = \frac{1}{\left\|\begin{bmatrix}1\\0\\1\end{bmatrix}\right\|}\begin{bmatrix}1\\0\\1\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\1\end{bmatrix}$.

$$u_2 = \cdots = \frac{1}{\sqrt{18}}\begin{bmatrix}-1\\4\\1\end{bmatrix}$$

$$u_3 = \cdots = \frac{1}{3}\begin{bmatrix}-2\\-1\\2\end{bmatrix}.$$

Let $P = \begin{bmatrix}u_1 & u_2 & u_3\end{bmatrix}$. Then

$$A = P\begin{bmatrix}7&0&0\\0&7&0\\0&0&2\end{bmatrix}P^T.$$

Let's go back and check that

$\left\{\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}-1\\4\\1\end{bmatrix}, \begin{bmatrix}-2\\-1\\2\end{bmatrix}\right\}$ is truly orthogonal.

$\left(\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}-1\\4\\1\end{bmatrix}\right) = 0$ (by Gram-Schmidt)

Since $\begin{bmatrix}-2\\-1\\2\end{bmatrix}$ is an eigenvector for eigenvalue 2

and $\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}1\\4\\1\end{bmatrix}$ are eigenvectors for eigenvalue 7

Thm 1 tells us that $\left(\begin{bmatrix}-2\\-1\\2\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix}\right) = 0$ and

$$\left(\begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix} \right) = 0.$$

Theorem 3 [Spectral Theorem Part II].

Suppose $A \in M_{n \times n}$ is symmetric.

Then (a) A has n real eigenvalues counting multiplicities.

i.e. $\det(A - xI) = (\lambda_1 - x)^{n_1} \cdots (\lambda_k - x)^{n_k}$

$\lambda_1, \ldots, \lambda_k \in \mathbb{R}$ are <u>distinct</u>.

(b) $\dim \operatorname{Nul}(A - \lambda_j I) = n_j \quad \forall\ 1 \le j \le k$.

Sketch of proof. Thm 7 (b) in Section 5.3.

<u>Spectral decomposition</u>

Suppose $A \in M_{n \times n}$ is symmetric

and $A = PDP^T$ with $P = [u_1 \cdots u_n]$

and $\{u_1, \ldots, u_n\}$ an orthonormal basis of $\mathbb{R}^n$ of eigenvectors of A.

If $D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$. Then $A u_j = \lambda_j u_j \quad 1 \le j \le n$.

Then

$$\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \begin{bmatrix} u^T \end{bmatrix}$$

$$A = \begin{bmatrix} u_1 & \cdots & u_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \begin{bmatrix} u_1^T \\ \vdots \\ u_n^T \end{bmatrix}$$

$$= \begin{bmatrix} \lambda_1 u_1 & \cdots & \lambda_n u_n \end{bmatrix} \begin{bmatrix} u_1^T \\ \vdots \\ u_n^T \end{bmatrix}$$ (matrix mult.)

$$= \underline{\lambda_1 u_1 u_1^T + \lambda_2 u_2 u_2^T + \cdots + \lambda_n u_n u_n^T}$$

Spectral decomposition

Notice that $u_j u_j^T \in M_{n \times n}$. and

that $(u_j u_j^T)^T = (u_j^T)^T u_j^T = u_j u_j^T$

i.e. $u_j u_j^T$ is symmetric.

Notice also that

$$Av = (\lambda_1 u_1 u_1^T + \cdots + \lambda_n u_n u_n^T) v$$

$$= \lambda_1 u_1 u_1^T v + \cdots + \lambda_n u_n u_n^T v$$

$$= \lambda_1 \langle v, u_1 \rangle u_1 + \cdots + \lambda_n \langle v, u_n \rangle u_n$$

$$= \lambda_1 \operatorname{proj}_{u_1} v + \cdots + \lambda_n \operatorname{proj}_{u_n} v.$$